

Patient's understanding of and satisfaction with acute stroke treatment

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ABSTRACT

Objective: We aimed to assess patient and surrogate understanding of and satisfaction with communication regarding acute stroke treatments of intravenous thrombolysis (IVT) and endovascular therapy (EVT).

Methods: In this single health-system prospective observational study, patients or their surrogates were interviewed within 72 h of acute stroke therapy. Respondent's satisfaction and self-reported understanding were rated on a Likert scale. Responses to open-ended questions were evaluated for recall of purpose and risks of treatment.

Results: Of 56 completed interviews (24 patients and 32 surrogates), 33 patients received IVT alone, 11 IVT and EVT, 12 EVT alone. Forty participants (71%) reported being extremely satisfied with their acute stroke care, 46 (82%) reported no difficulty understanding the purpose of treatment, while 36 (64%) reported no difficulty understanding risks. Two or more risks were verbalized by 8 (24%) participants for IVT, 2 (17%) for EVT, and 7 (64%) for both IVT and EVT. Brain bleeding was the most recalled risk for IVT and "lack of benefit" for EVT.

Conclusions: Majority of the participants were extremely satisfied and reported no difficulty understanding purpose and risks of acute stroke treatment, however only 30% were able to verbalize two or more risks associated with the treatment.

1. Introduction

In treatment of acute ischemic stroke, communication regarding, benefits, risks and alternatives of acute stroke treatments is done in a time-efficient manner. Decisions are made with consent or assent from patient or surrogate, or at times with physician documentation to deliver standard of care treatment as quickly as possible. Recent professional society consensus statement on the consent discusses understanding, appreciation, reasoning, and choice as elements of an informed consent [1]. If patients are treated on an emergent basis, patient or surrogate should be informed later. In clinical practice, variability exists, with consent reported to occur in less than a minute in majority of cases [2]. The expectations for the high-quality consent appear to vary between clinicians and persons of non-medical or legal background [3]. The perspective of the patient, or the decision-making surrogate, arguably the center of the consenting process, has not been previously studied. Understanding the patient's perspective is crucial to design an efficient

decision-making process that incorporates the needs of the persons suffering acute stroke. We aimed to assess patient and surrogate's understanding of and satisfaction with communication regarding acute stroke reperfusion therapy, namely intravenous thrombolysis (IVT) and endovascular therapy (EVT), delivered during telestroke or in-person encounters.

2. Methods

We conducted prospective observational study at a comprehensive stroke center and telestroke sites within a health system in Northeast Ohio between November, 2020 and May, 2021. Patients or surrogate decision makers who were offered IVT or EVT and who were 18 years or older were included. All acute thrombolysis decisions involved one of 10 vascular neurologists or 3 vascular neurology fellows and endovascular therapy involved vascular neurologists or vascular neurology fellow and one of the 6 neuroendovascular attending physicians (neurologist,

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neurosurgeon, or neuroradiologist) and 3 neuroendovascular fellows. We excluded patients with aphasia or insurmountable communication barriers or cognitive impairment or intoxication, with no surrogate decision makers at the time of consent. A structured interview with open-ended and multiple-choice questions was conducted over telephone or in person within 72 h of receiving acute stroke treatment. Respondents' satisfaction and self-reported understanding were rated on a Likert scale. Transcribed responses to open-ended questions were evaluated for recall of the diagnosis, treatment, purpose and risks of the treatment by two blinded reviewers and disagreements were resolved by discussions during research team meetings. Health literacy was assessed by a validated single item health literacy screener with a 5-point Likert scale response: "How often do you need to have someone help you when you read instructions, pamphlets, or other written material from your doctor or pharmacist?" [4] If participants were unavailable, we attempted to reach them repeatedly, at least 2 h later. Complications within the first 24 + /- 12 h were abstracted; early neurologic deterioration (END) was defined as worsening of NIH Stroke Scale score by ≥ 4 points, including with presence of intracranial hemorrhage. The study was judged as minimal risk and therefore was exempt from the regular Institutional Review Board review. We used descriptive statistics, median and interquartile range (IQR), unless stated otherwise.

3. Results

Between November 2020 and May 2021, 128 acute stroke presentations met inclusion criteria, of which 54 were unavailable to answer at least two phone calls done two hours apart, 15 patients/surrogates refused interview and 3 interviews were incomplete. Fifty six interviews were completed, 24 by patients and 32 by surrogates (Fig. 1). Thirty three (59%) patients received IVT, 11 (20%) received both IVT and EVT, and 12 (21%) received EVT alone (Table 1). Median NIH stroke score of patients was 5 (IQR: 3–9). When assessed for health literacy, 53 out of 56 participants "never" or "rarely" need help with information from a doctor or pharmacist.

Forty participants (71%) reported being extremely satisfied with their acute stroke care, 46 (82%) reported no difficulty understanding the purpose of treatment, while 36 (64%) reported no difficulty understanding risks (Table 2). When asked for spontaneous comments, 5 participants (9%) described stressful emotions, 5 (9%) admitted limited understanding, and 3 (5%) considered the information on risks overly detailed (Supplemental content, Appendix).

Table 3 details objective patient's understanding. All 56 participants (100%) understood their diagnosis, 39 (89%) the purpose of IVT, and 21

Table 1

Participant characteristics.

Respondent Age in years,* median (IQR)	62 (52-70)
Respondent level of education, n (%)	
• Less than high school	6 (11%)
• High school graduate	18 (32%)
• College graduate	16 (29%)
• Graduate school	13 (23%)
• Undisclosed / missing	3 (5%)
Patient's Age in years, median (IQR)	73 (61-80)
Patient's Race, n (%)	
• Black	13 (23%)
• White	42 (75%)
• Asian	1 (2%)
Treatment received	
• IVT	33 (59%)
• IVT and EVT	11 (20%)
• EVT	12 (21%)
Initial NIHSS, median (IQR)**	5 (3-9)
NIHSS score at 24 h + /- 12 h**	4 (1-12)
Treatment complications, n (%)	12 (21%)
• Early neurological deterioration**	8 (14%)
• Symptomatic intracranial hemorrhage	0
• Bodily bleeding (e.g. hematuria)	3 (5%)
• Failure of recanalization during EVT	2 (4%)
• Arterial perforation during EVT	1 (2%)
• Other ^a	2 (4%)
Treatment consenting mode, ^b n (%)	
IVT	
• Telephone	1
• Telestroke	42
• In-person	5
EVT	
• Telephone	15
• Telestroke	10
• In-person	6

IVT: Intravenous thrombolysis; EVT: Endovascular Treatment, NIHSS: National Institute of Health Stroke Scale

* Missing data from one respondent

** Missing data from five patients

^a one patient with tooth fracture, one with IV extravasation

^b Responses in the survey question are not mutually exclusive, some of the patients received information using two different modes.

(91%) for EVT. While 51 participants (91%) spontaneously named at least one risk of treatment correctly, only 17 (30%) were able to name two or more risks. Most recalled risks were brain and bodily bleeding for IVT, and lack of benefit for EVT (Fig. 2). Six participants reported complications with treatment, 2 with IVT only, 2 with IVT and EVT, 2 with EVT only.

We explored factors associated with patient's satisfaction and self-reported understanding in a post-hoc analysis. Composite of participants' satisfaction negatively correlated with composite of self-reported difficulty in understanding ($p < 0.01$, Fig. 3), and with presence of treatment complication (non-significant after correction for multiple testing), but not with the number of risks recalled, health literacy, clinical improvement at 24 h (Supplemental Table 1). Composite of patients' understanding trended towards negative correlation with health literacy and with self-reported overwhelming feeling during consent, but findings lost statistical significance after Bonferroni correction for multiple testing (Supplemental Table 2).

4. Discussion

Our study explored patients' and their surrogates' satisfaction with, knowledge of, and perspectives on communication for acute reperfusion treatment. Majority of participants were satisfied with the information they received on the acute stroke treatments, reported no difficulty in understanding the purpose and risks, and were able to name one potential risk of treatment. But a quarter of patients found the information difficult to understand "moderately" or "quite a bit" and a smaller

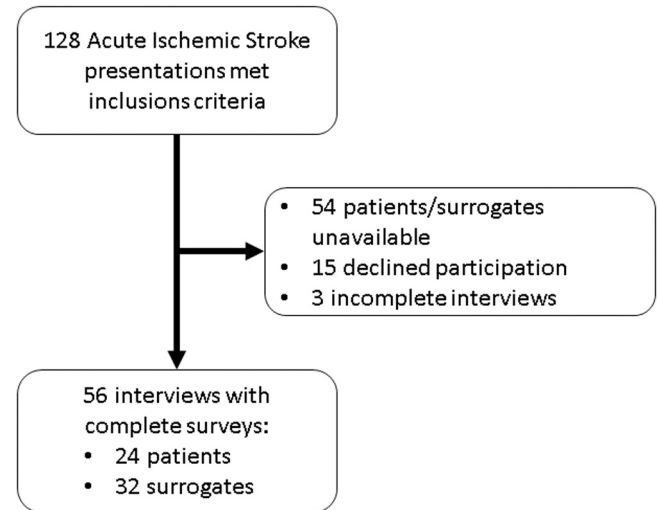


Fig. 1. Flow Chart of Study cohort.

Table 2
Likert scale rating of participant satisfaction and self-reported understanding.

Questions On Understanding Of Acute Stroke Treatment					
How difficult was it for you to understand ...	Not at All	A Little	Moderately	Quite a Bit	Extremely
... the information that you received about your (your family member's) stroke?	28 (50%)	14 (25%)	13 (23%)	1 (2%)	0
... what was done for your (your family member's) stroke at the emergency room?	36 (64%)	11 (20%)	7 (12%)	2 (4%)	0
... what was the purpose of the clot-dissolving/clot-removing therapy that you (your family member) were offered?	46 (82%)	5 (9%)	3 (5%)	1 (2%)	1 (2%)
... what were the risks of the clot-dissolving/clot-removing therapy that you (your family member) were offered?	36 (64%)	13 (23%)	4 (7%)	1 (2%)	2 (4%)
Questions On Satisfaction With Acute Stroke Treatment					
How satisfied are you with the care you received at the acute stroke care unit?	0	0	2 (4%)	11 (20%)	40 (71%)
How would you rate your satisfaction with the information that you received about ...	Poor	Fair	Good	Very Good	Excellent
... the treatment (clot-dissolving/clot-removing)?	1 (2%)	1 (2%)	8 (14%)	23 (41%)	23 (41%)
... the purpose/benefits of the treatment?	1 (2%)	1 (2%)	9 (16%)	20 (36%)	25 (44%)
... the risks of the treatment?	2 (4%)	5 (9%)	6 (11%)	22 (39%)	21 (37%)

Table 3
Evaluation of transcribed responses for recall of diagnosis, purpose, and risks of treatments.

Evaluated Criteria	n (%)
Recalled the diagnosis	56 (100%)
Recalled treatment	
• IVT	42 (95%)
• EVT	23 (96%)
Recalled purpose of treatment	
• IVT	39 (89%)
• EVT	21 (91%)
Recalled at least one risk involved with treatment	
• IVT	29 (88%)
• IVT and EVT	11 (100%)
• EVT only	11 (92%)
Recalled two or more risks involved with treatment	
• IVT	8 (24%)
• IVT and EVT	7 (64%)
• EVT only	2 (17%)

proportion were “a little” or “not at all” satisfied with the information on risk of treatment. The low proportion of patients who recalled a second risk could be caused by poor retention or limited focus of the patient or surrogate, or inconsistency or lack of emphasis of the clinicians. Recently, American Academy of Neurology suggested patients or surrogates with decision-making capacity “should be quickly but clearly

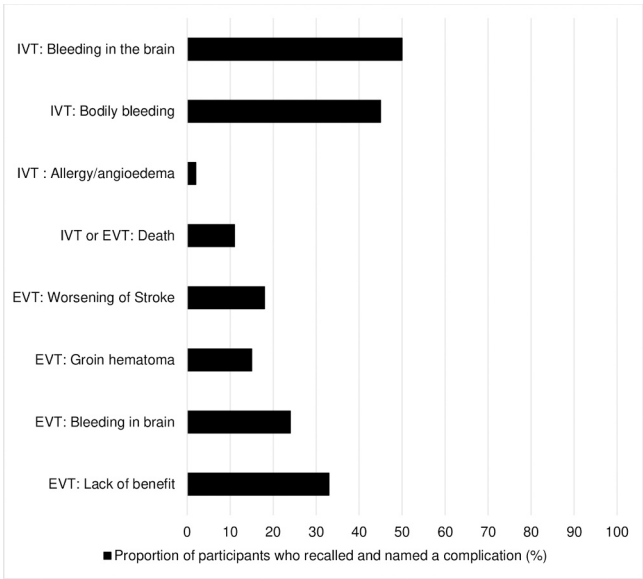


Fig. 2. Objective evaluation of transcribed responses for recalled risks of intravenous thrombolysis (IVT) and endovascular therapy (EVT), (n = 51).

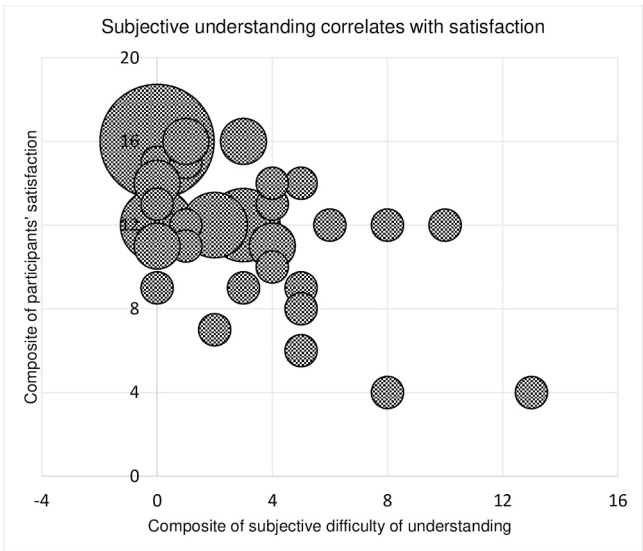


Fig. 3. Difficulty of understanding negatively correlates with satisfaction. Composite of self-reported difficulty of understanding acute stroke treatment (X axis, higher score indicates worse understanding) negatively correlates with composite of participant satisfaction with information received on acute stroke care (Y axis, higher score indicates better satisfaction), Spearman's $\rho = -0.52$ ($p < 0.01$, after Bonferroni correction for 24 correlations). Size of each bubble is proportional to the number of participants.

informed” of the risk of intracranial hemorrhage and angioedema [1]. Our observation of recall of angioedema in only one out of 44 participants treated with IVT may suggest that in practice, angioedema is not understood or forgotten or providers are not communicating consistently or effectively. We observed good self-reported understanding and participants’ satisfaction despite limited recall of risks. Good to excellent satisfaction with the information received about the risks of 88% might mean that they do not want the details of the risks of treatment. Yet, 12% had poor or fair satisfaction of risks information, numerically greater proportion than 4% that reported poor or fair satisfaction with treatment or benefits of treatment information. Future research should examine the impact of visual aids on patients’ comprehension,

experience, and recall [5].

Several limitations may have affected our observations. The participants represent a sample and might not represent patients and surrogates of those who undergo acute stroke therapy. As a single health system study of 56 patients, we could not explore the differences by racial or socioeconomic subgroups, settings (telemedicine or in-person, type of hospital), treatment modality, or providers. Information is provided in ongoing overlapping fashion by multiple providers and settings, and it may be impossible to tease apart providers and modes of information delivery. Our study may have low power to evaluate effect of patient outcome or treatment complications on responses.

5. Conclusion

Most patients and surrogates are satisfied with the information they have received regarding acute stroke treatment. Despite the majority reporting no difficulty in understanding the risks of acute stroke treatments, only 30% were able to recall and verbalize two or more risks.

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Vanguru Husitha: Conceptualization, Data curation, Investigation, Methodology, Writing – original draft. **Uchino Ken:** Conceptualization, Data curation, Formal analysis, Methodology, Project administration, Supervision, Writing – review & editing. **Blaginykh Elena:** Data curation, Formal analysis, Investigation, Methodology, Project administration, Writing – review & editing. **Handshoe Lacy Samuel:** Data

curation, Investigation, Methodology, Writing – review & editing. **Čierny Marek:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – review & editing.

Declaration of Competing Interest

KU has received compensation from Abbott Laboratories and Genentech, Inc.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.clineuro.2024.108163](https://doi.org/10.1016/j.clineuro.2024.108163).

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